GEA1000 – Quantitative Reasoning with Data Project
AY2025/2026 Semester Two

NUS Health & Wellbeing Interns

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Tutorial E38 Group 1

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Part A

Chosen Topic for Part A: Sleep quality of Singapore residents: findings from the 2016 Singapore mental health study [1].

Section 1

1. Aim of Study

The primary objective of this study was to investigate the quality of sleep among community dwellers in Singapore. Specifically, the researchers wanted to:

1. Characterise the sleep quality of adult Singaporean residents in a nationally representative sample.
2. Identify the sociodemographic factors that correlate with poor sleep in this population. The sociodemographic correlates in the study include Gender, Ethnicity, Marital status, Education, Lifestyle (smoking, drinking, etc.), Mental health illnesses, and physical health illnesses.

2. Main Independent and Dependent Variables

Dependent Variables:
Sleep Quality: Measured using the Pittsburgh Sleep Quality Index (PSQI), a self-report questionnaire that assesses sleep patterns over a one-month interval.

The PSQI is a standardized clinical tool that asks about seven components: subjective sleep quality, sleep latency (time to fall asleep), sleep duration, habitual sleep efficiency, sleep disturbances, use of sleeping medication, and daytime functioning.

A global score was calculated from 0–21; a total score of 5 or more was used to define "poor sleepers," while a score below 5 indicated "good sleepers." Sleep duration obtained was then classified into ≤ 6h, 7–8h and ≥ 9h.

Independent Variables:

- *Mental Disorders:* Measured using the World Health Organization Composite International Diagnostic Interview (WHO CIDI 3.0) to diagnose conditions such as Major Depressive Disorder and Obsessive-Compulsive Disorder. Respondents were categorized into three types: no mental illness, at least one mental illness, and two or more comorbid mental illnesses in the past 12 months.
- *Physical Disorders:* Measured by participant self-report of chronic medical conditions (e.g. diabetes, heart disease) that had been diagnosed by a doctor. Responses to the checklist were grouped into 3 types: no chronic physical conditions, at least one physical chronic condition, and multimorbidity in the past 12 months. Participants were given a list of 18 common chronic medical conditions in Singapore such as hypertension, diabetes, asthma, and neurological disorders and were asked if a healthcare professional had ever told them they had the condition.

The researchers state that the observed stronger and more consistent associations between sleep and the above two independent variables. Hence, we believe that mental and physical disorders were the main 2 independent variables, while the other sociodemographic factors were considered secondary independent variables.

3. Main Findings of the Article

a) "In this sample, only certain measures of socioeconomic status were moderately associated with sleep quality and duration. Instead, we observed stronger and more consistent associations between sleep and physical and mental health conditions" [1].

b) The target population were residents of Singapore who were adults (including citizens and permanent residents) aged 18 years and above. The study used disproportionate stratified sampling across three main ethnic groups (Chinese, Malay, Indian) and oversampled those aged 65 and above, with sampling weights then applied to make the findings generalisable to the broader adult Singapore population.

c) To examine whether mental health conditions are significantly associated with poor sleep quality among Singapore residents, the following hypotheses were formulated:

   H_0: There is no significant association between mental health conditions and poor sleep quality among Singapore residents.

   H_1: There is a significant association between mental health conditions and poor sleep quality among Singapore residents.

   The results of the chi-square, logistic regression and multimodal analysis tests revealed a statistically significant association between mental health conditions and poor sleep quality. Relative to individuals with no mental illness, those diagnosed with one mental illness demonstrated significantly higher odds of reporting poor sleep quality (AOR $= 3.42$, 95% CI $[2.36,\ 4.95]$, $p < 0.001$). Furthermore, individuals with comorbid mental illnesses exhibited a substantially greater likelihood of poor sleep quality (AOR $= 14.11$, 95% CI $[6.52,\ 30.54]$, $p < 0.001$), suggesting a dose-dependent relationship between the number of mental health conditions and the severity of sleep impairment.

   Decision: As both p-values were well below the predetermined significance level of $\alpha = 0.05$, and neither confidence interval contained the value of 1, there was sufficient statistical evidence to reject the null hypothesis. The findings indicate that mental health conditions are significantly and positively associated with poor sleep quality among Singapore residents.

Section 2

4. Confounders in the Observational Study

This was an observational study that aimed to identify socio-demographic factors correlated with sleep quality. The researchers also included clinical health issues and smoking and drinking habits as areas to look into. These qualities are considered independent variables for the study. There were no specific confounders controlled by the researchers, but these independent variables could serve as confounders for each other. The confounding independent variables that are controlled by the researchers include mental illnesses, chronic physical illnesses, ethnicity, gender, education level, household income, age, smoking habits, drinking habits, marital status, employment status and body mass index.

Gender is a confounding variable. Being female was associated with both higher occurrence of mental illness and lower sleep quality, thus gender may confound the association between mental illness and sleep quality.

A potential confounder not controlled by researchers is whether the participant's house is located near sources of nighttime noise, such as busy roads or areas with active nightlife. Loud noises may disrupt sleep, leading to poor sleep quality.

5. Selection Process

The subjects were Singapore citizens and permanent residents over the age of 18. There were 6,126 subjects with a response rate of 69.5%. This sampling frame covers the general population of Singapore residents fairly well. However, it excludes children younger than 18 years old, residents unable to attend the face-to-face interview due to being out of the country, being institutionalized or hospitalized, residents unable to complete the interview due to physical and/or mental illnesses, language barriers, and residents that could not be contacted using information from the national registry. Probability sampling was employed in the selection of respondents.

Due to its failure to account for certain groups of the target population, the study may lack crucial information from these non-responding or excluded subjects. Some groups of the population may have a rarer occurrence, such as chronic physical illness. With a sample size of 6,126, this may cause these smaller subgroups to have less or even no representation in the study, leading to insufficient evidence for reliable results in these groups. Thus, this study cannot be fully generalized to the target population of Singapore residents.

6. Concluding Evaluation

This study put in good efforts in conducting their study well. The researchers did their best to minimise non-response through repeated attempts at contacting selected respondents at different timings. The study also uses disproportionate stratified sampling to select more subjects from the elderly and minority ethnicities, then applies sampling weights to adjust the results to represent Singapore's population. As such, this allowed enough data of these groups with lesser numbers to be collected for more reliable results.

However, the study had some weaknesses. It relied on participants to self-report their quality of sleep through the Pittsburgh Sleep Quality Index. While this method was chosen for its feasibility, participants may misrecall their sleep characteristics or be affected by recency or other biases, causing inaccurate reporting of their sleep quality and some unreliability. The data were also collected from 2016 to 2018, making them relatively old compared to the current 2026 context.

The study is generally well-conducted. The strengths in having a relevant sampling frame, random sampling, and stratified sampling of some minority groups outweigh the weaknesses of the study method, older data and lack of generalisability for smaller subgroups. By having a sample representative of the general population rather than a specific demographic, it provides insights into the sleep quality of Singapore residents.

Part B

Section 1: Understanding Data and Data Cleaning

1. Data Types

The dataset contains 1,500 observations. For the analysis used in Parts B and C, their types are as follows:

Name	Type	Remark
Country	Categorical (nominal)	Singapore, Malaysia, Indonesia
PA_Days	Numerical (discrete)	0–7
VA_Weekly_Days	Numerical (discrete)	0–7
VA_Daily_Hours	Numerical (continuous)	
MIA_Weekly_Days	Numerical (discrete)	0–7
MIA_Daily_Hours	Numerical (continuous)	
Hours_Slept	Numerical (discrete)	
Fruits	Numerical (discrete)	1–8
Relaxedness	Categorical (ordinal)	1–5
University	Categorical (nominal)	NIE, UPM, UM, UI, UGM, UA
Gender	Categorical (nominal)	M, F
BMI	Numerical (continuous)	

Table 1: Data types for variables used in the analysis.

2. Data Cleaning

Inspecting the dataset, we observe that there were several unusual observations and the following cleaning steps were applied:

- **Inconsistent country and university naming.** `Country` uses inconsistent capitalisation: "Singapore" and "Indonesia" are title-cased, while "MALAYSIA" is fully capitalised. Similarly, `University` names are stored as full formal names. These are standardised during preprocessing: country names are mapped to ISO codes (SG, MY, ID) and university names are abbreviated (e.g., "Universitas Indonesia" $\rightarrow$ "UI") for clarity.

- **GPA values are recorded on different scales.** Singapore universities appear to use a 5-points GPA scale while the other universities use a 4-point scale. This was substantiated through three lines of evidence: a frequency count showed over 1,000 students scoring between 0-4 with only a single observation above 4, a histogram of GPA scores showed a distribution that naturally peaked and terminated at 4, and cross-referencing the actual universities listed in the dataset confirmed that their institutional GPA scales are out of 4. To make GPA comparable across countries, 1.0 is subtracted from all Singaporean GPA values to bring the onto the same 4-point scale. Following this adjustment, 1 observation with a GPA of 4.7 is dropped, as it remains outside the valid range and is likely an error.

- **BMI values given as NA.** There are 102 entries with "NA" in the BMI column, which have been removed.

- **BMI values of zero.** 16 observations have a BMI of 0, which is impossible, and have been removed.
- **Hours Slept outliers.** 12 observations were outliers for `Hours_Slept` and have been removed.

This yields a final cleaned dataset of 1,369 observations. Removing these observations, rather than imputing, is justified because the affected rows represent a small fraction of the total dataset (less than 10%), meaning their removal is unlikely to introduce meaningful bias.

Section 2: Data Visualisation and Analysis

3. PA Days Analysis

An appropriate visualisation of the distribution of `PA_Days` is a bar chart, given in Figure 1, since `PA_Days` is a discrete numerical variable that takes integer values from 0 to 7. The five-number summary, mean, and standard deviation for `PA_Days` are: mean $= 1.80$, std $= 1.76$, min $= 0.00$, Q1 $= 0.00$, median $= 1.00$, Q3 $= 3.00$, max $= 7.00$.

The data on `PA_Days` suggest that physical activity levels among participants are quite low. The median is just 1 day per week, meaning at least half of the participants engage in 30+ minutes of physical activity on only one day or fewer per week. The Q1 of 0 indicates that at least 25% of respondents reported no physical activity at all. The distribution is right-skewed, with most values concentrated at the lower end (0–2 days). The World Health Organisation (WHO) recommends that adults engage in at least 150–300 minutes of moderate-intensity aerobic activity per week, which, at 30 minutes per session, would correspond to at least 5 days per week. Very few participants meet this threshold, suggesting that most respondents in this dataset fall well below recommended physical activity levels.

4. BMI Analysis

`BMI` is transformed into the categorical variable `BMI_Class` using the Asian BMI cut-off points.

From Figure 2, we observe that Singapore stands out with a markedly lower proportion of underweight students (only 3.6% versus $\sim$20% for the other two countries), and a notably higher proportion of overweight and obese students (combined 47.3%). Malaysia has the highest proportion of underweight students (23.4%). Indonesia's distribution is relatively balanced between categories. These differences may reflect dietary patterns, lifestyle habits, or compositional differences in the samples from each country, although the relatively small sample sizes from Singapore and Malaysia mean that these country-level comparisons should be taken with caution.

5. Physical Activity vs Body Weight Analysis

`MET_Daily_Hours` is a composite measure of physical activity that accounts for both the intensity and volume of activity by weighting each activity type by its metabolic equivalent: moderate intensity (4.0 METs) and vigorous intensity (8.0 METs), then multiplying by duration and frequency. This makes it a more precise and meaningful measure compared to simply counting physical activity days. This engineered feature is based on the MET minutes/week measure from the IPAQ report [2].

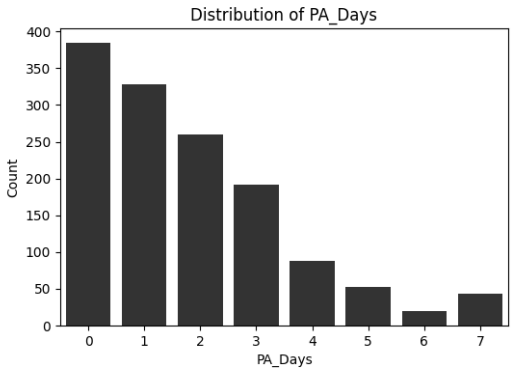

Figure 1: Bar chart showing the distribution of `PA_Days` among survey participants.

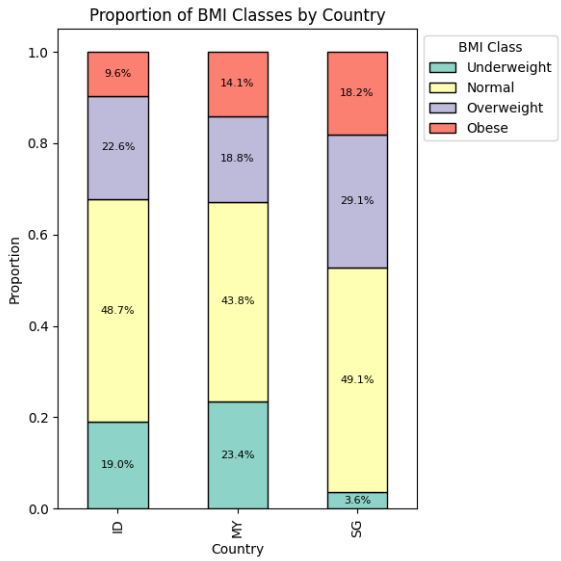

Figure 2: Stacked percentage bar chart comparing the proportions of BMI class between countries.

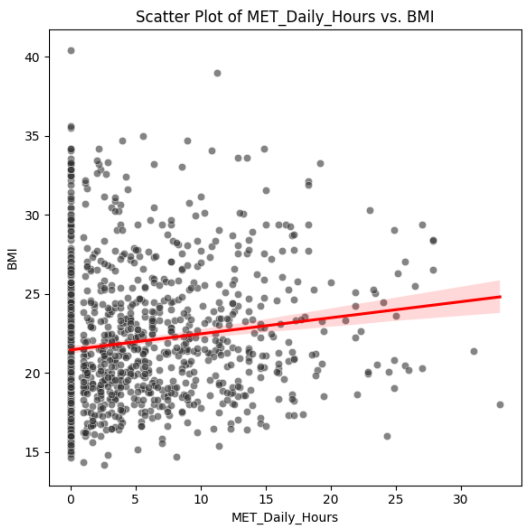

Figure 3: Scatter plot of `MET_PA_hours` against `BMI`, with a fitted regression line.

The scatter plot of `MET_Daily_hours` versus `BMI` in Figure 3 shows a weak positive correlation ($r \approx 0.15$). Contrary to the expectation that more physical activity would correlate with lower BMI, the relationship here is positive, though the association is very weak and not practically meaningful. This may be due to the cross-sectional nature of the survey (it captures a single point in time rather than long-term habits) and the fact that BMI is influenced by many other factors not captured here (e.g., diet, genetics).

Figure 4 shows that for both Singapore and Indonesia, `MET_Daily_Hours` and `BMI` have weak positive correlations, consistent with the overall dataset trend. However, in Malaysia, there is little to no correlation ($r = 0.04$), suggesting that unlike Singapore ($r = 0.14$) and Indonesia ($r = 0.15$), physical activity does not show any observable relationship with BMI in this sample.

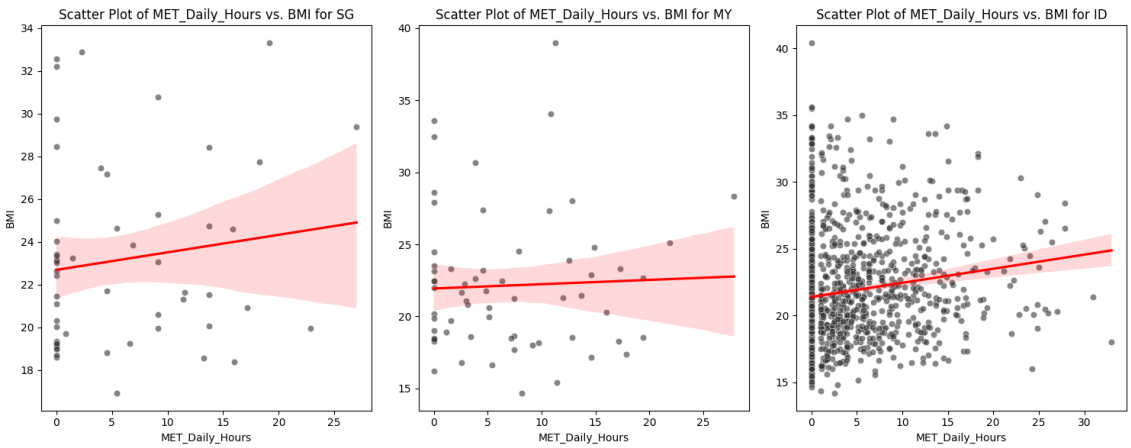

Figure 4: Scatter plots of `MET_Daily_Hours` against `BMI`, stratified by country (Singapore, Malaysia, Indonesia), each with a fitted regression line.

The overall weak relationships suggest that physical activity as measured here is not a strong predictor of BMI across this population.

6. Testing the Dean's Suspicions: Physical Activity

We use the MET (Metabolic Equivalent of Task) physical activity category as the variable to represent levels of physical activity. The criterion for categorization is based on MET minutes/week and several other PA variables. The resulting MET category classifies participants into three groups: "inactive", "minimally active", and "HEPA active", which are recoded as a binary variable: High PA ("HEPA active") and Low PA ("inactive" and "minimally active").

Relaxedness is recoded as: Relaxed (Relaxedness score of 4 or 5, i.e., feeling relaxed "Often" or "All of the time") and Not Relaxed (score of 1–3).

Group	PA Level	Not Relaxed	Relaxed	% Relaxed
Overall	High PA	123	127	50.8%
	Low PA	127	592	52.9%
Male	High PA	64	44	40.7%
	Low PA	377	405	51.8%
Female	High PA	59	83	58.5%
	Low PA	150	187	55.5%

Table 2: Relaxedness by PA level, overall and stratified by gender.

Overall, the association between PA level and relaxedness is very weak: High PA students are marginally less likely to feel Relaxed (50.8%) than Low PA students (52.9%), a difference of just 2.1 percentage points that does not support the Dean's claim.

However, the gender-stratified results in Table 2 reveal a striking reversal. Among males, High PA is negatively associated with relaxedness (40.7% vs 51.8%, a difference of 11.1 percentage points), while among females the direction flips, with High PA females marginally more likely to feel Relaxed (58.5% vs 55.5%). This reversal is consistent with Simpson's Paradox, and confirms that gender is a confounder: it is associated with both PA level and relaxedness,

and controlling for it materially changes the observed association. The Dean's claim therefore only partially holds: the positive association between physical activity and relaxedness exists weakly among females but reverses among males.

7. Dean's Suspicion about Fruit Consumption among Singaporean Students

The Dean's hypothesis is that more than half of all Singaporean university students consume fewer than 2 fruits per day, which is below the Health Promotion Board (HPB's) recommendation. The null and alternative hypotheses are:

$$H_0 : p = 0.5,$$
$$H_1 : p > 0.5$$

Among the 55 Singaporean students in the cleaned dataset, 27 (49.1%) consumed fewer than 2 servings of fruit per day. To test the probability of observing this sample proportion or a more extreme statistic by chance, a one-tailed binomial test was conducted, yielding a p-value of 0.606. Therefore, we fail to reject H_0 at the 5% significance level. There is insufficient evidence in this dataset to confirm the Dean's suspicion.

The observed proportion is slightly below 50%, pointing against the Dean's claim rather than supporting it. However, there are several limitations that must be acknowledged: the sample of Singaporean university students was not specifically designed to be representative of the population of all Singaporean university students. In addition, the sample size is very small and may be insufficient to detect a small but meaningful difference from the hypothesised proportion of 0.50, potentially resulting in low statistical power. Furthermore, fruit consumption was self-reported from the survey, which may introduce recall bias or social desirability bias (overreporting of healthy eating habits); respondents may also have varying interpretations of what constitutes one serving of fruit. Thus, while the test does not support the Dean's suspicion at this significance level, the findings should be interpreted with caution. Future study may benefit from replication using a larger, more representative sample.

Section 3: Thinking with Data and Reflecting on Data

8. Additional Analysis

Sleep is categorized into three groups: short sleep ($\leq$ 6h, $n = 397$), optimal sleep (7–8h, $n = 471$), and long sleep ($\geq$ 9h, $n = 501$).

Students in the ≤ 6 hours sleep category recorded the highest median sugary drink consumption (≈ 4 drinks) with the widest variability (IQR: 2–6), while both the 7–8 hours and ≥ 9 hours groups showed considerably lower medians of approximately 2 drinks and tighter distributions. This descending trend suggests that shorter sleep duration is associated with greater sugary drink consumption among Singapore students. A Chi-Square test of independence confirmed this association to be statistically significant ($\chi^2 = 27.71$, $p = 0.0156$), as the p-value falls below the $\alpha = 0.05$ threshold, thereby rejecting the null hypothesis that sleep duration and sugary drink consumption are independent variables.

`BMI_Class` and sleep quality also displayed a statistically significant association, with higher BMI categories linked to poorer sleep. This suggests that weight management could be an important lever for improving sleep among university students, and points to a potential policy intervention targeting physical activity and dietary habits together with the aim of reducing excess weight and, in turn, promoting better sleep outcomes.

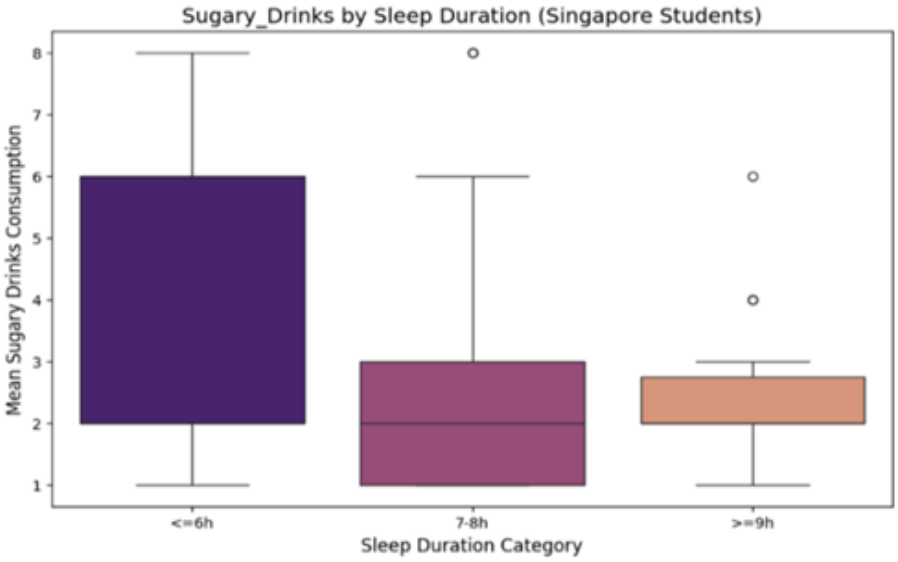

Figure 5: Box plot of sugary drink consumption across sleep duration categories.

Regarding fruit consumption, students consuming 2 or more servings reported slightly higher proportions of optimal sleep (41.4%) compared to those consuming fewer than 2 servings (37.7%). However, this difference is not statistically significant ($\chi^2 = 1.76$, $p = 0.184$), so we fail to reject the null hypothesis of independence. There is insufficient evidence to conclude that fruit consumption is associated with sleep quality.

9. Proposed Variable to Collect

We would propose the Pittsburgh Sleep Quality Index (PSQI) [3] to measure the sleep quality of university students. The PSQI is a validated self-report tool measuring overall sleep quality across 7 components, making it directly relevant to university student health. It asks respondents to reflect on their sleep habits over the past month. While the existing dataset captures sleep duration, this alone is insufficient. Two students sleeping the same number of hours may have vastly different sleep quality due to differences in sleep efficiency, disturbances, or daytime dysfunction. The PSQI addresses this gap by measuring sleep as a multidimensional construct, accounting for individual variation in sleep needs and capturing aspects of sleep health that duration simply cannot reflect.

Measurement. Students complete 19 items covering sleep duration, efficiency, disturbances, and daytime dysfunction, producing a global score from 0–21 (above 5 = poor sleep quality). It takes 5–10 minutes and can be embedded directly into a survey, requiring no external monitoring devices.

Limitations. As a self-report measure, PSQI responses are susceptible to recall and social desirability bias, where students may underreport sleep problems or misremember sleep patterns from the past month. Additionally, the PSQI captures average patterns over 30 days, so it may not reflect acute disruptions like exam periods, which are especially relevant for university students.

10. Evaluation of Data Analysis (Part B)

Aspect 1: MET Activity.

The use of MET-minutes per week as the primary measure of physical activity significantly adds value to the understanding of the relationship between physical activity and sleep duration. Rather than treating all active days as equal, MET-minutes captures both the intensity and volume of activity by applying weighted MET values. This allows for more accurate PA Level classification, which is more meaningful in the context of sleep research, as it enables the analysis to detect whether it is the degree of physical exertion, and not the frequency of activity, that is associated with differences in sleep duration.

A limitation of this analysis is that the physical activity data is derived from the IPAQ short-form questionnaire, which relies entirely on self-reporting. Respondents are required to recall the frequency and duration of their physical activities over the past week, which is susceptible to recall bias and social desirability bias, where participants may overestimate their activity levels.

Aspect 2: Generalisability of Findings.

In Part B, apart from Questions 7 and 8, the full sample was used for the analysis. However, the dataset is heavily skewed towards Indonesian students, who account for approximately 89% of the cleaned sample, while Singapore contributes only around 4%. This severe imbalance means that the overall findings are largely driven by the Indonesian subgroup and may not accurately reflect the health behaviours of NUS students specifically. Furthermore, the data were collected during the COVID-19 pandemic (March–June 2021), when movement restrictions and campus closures were in effect across all participating countries, likely suppressing typical physical activity levels and altering dietary and sleep habits. As such, there is a large discrepancy between the population of interest (NUS university students) and the sampling frame.

Appendix

All data analysis conducted as part of this study, including data preprocessing, statistical analyses, and figure generation, is documented in the project's code repository, available at https://github.com/chinzhening/GEA1000. The repository contains the raw and cleaned datasets (where permissible), annotated scripts, and instructions for reproducing all tables, charts, and summary statistics in this report. This ensures transparency, reproducibility, and allows other researchers or students to explore and build upon the work.

AI Tool Used	**Prompt and Output**	**How the output is used in the assignment**
Claude	Link	To clarify confusion, rewrite points and ask for feedback

Table 3: Declaration of AI use.

References

[1] Y. Y. Lee, J. H. Lau, J. A. Vaingankar, R. Sambasivam, S. Shafie, B. Y. Chua, W. L. Chow, E. Abdin, and M. Subramaniam, "Sleep quality of singapore residents: findings from the 2016 singapore mental health study," *Sleep Medicine X*, vol. 4, p. 100043, 2022.

[2] IPAQ Research Committee, "Guidelines for data processing and analysis of the international physical activity questionnaire (ipaq)," 2004, available online.

[3] D. Buysse, C. Reynolds, T. Monk, S. Berman, and D. Kupfer, "The pittsburgh sleep quality index (psqi): A new instrument for psychiatric research and practice," *Psychiatry Research*, vol. 28, no. 2, pp. 193–213, 1989. [Online]. Available: https://www.sleep.pitt.edu/sites/default/files/assets/Instrument%20Materials/PSQI-Article.pdf